JMB Lighting — Safety & Safe Operation Guide

JMB Zone / 4our / 8ight LED lighting controllers · Read before installation and keep for future reference. · 2026-09-16

> **DRAFT — legal review required.** Technical content reflects the shipped design. Passages marked **[LEGAL]** need wording confirmed or supplied by JMB Lighting's legal advisers before publication. Compliance markings (UKCA/CE, Bluetooth qualification) must be completed before sale.

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1. Intended use

JMB controllers drive low-voltage constant-voltage LED strip and fixtures (RGB, tuneable white, single-colour and infrared) for film, TV, broadcast, stage and architectural use, controlled by DMX512, CRMX wireless or the companion Bluetooth app. They are professional equipment intended for installation and operation by competent persons.

Indoor, dry locations only. The units are not sealed against water or dust and must not be used outdoors unprotected, in wet or condensing environments, or where flammable vapours are present.

Any other use — including driving non-LED loads, mains loads, or exceeding the ratings below — is outside the intended use. **[LEGAL: liability wording for misuse.]**

2. Electrical safety

- **JMB Zone, 4our and 8ight each contain a mains power supply.** Connect only to a properly earthed mains outlet of the voltage stated on the rating plate. Do not open the enclosure: no user-serviceable parts inside; hazardous voltages are present when connected. Disconnect from the mains before any cleaning, fuse replacement or cabling work.
- **Mains inlet.** The mains connection is a locking powerCON TRUE1 inlet on every unit. Use the supplied lead or one of the same rating; latch it fully before switching on, and never unplug it under load.
- Output circuits are separated extra-low voltage (12 or 24 V DC). Route output wiring away from mains cabling.
- Use conductor sizes rated for the configured port current (up to **6.75 A** per output on Zone units and 5 mΩ-sense modules, 6.5 A on earlier 4our/8ight modules). Undersized strip feed wiring is a fire risk that no controller protection can fully remove.

3. Fuses

Mains fuse (rear panel). The mains supply runs through a 5 × 20 mm ceramic cartridge fuse in the screw-cap holder on the rear panel. Unplug the power cable before opening the holder, and replace the fuse only with the same type and rating, which is also printed beside the holder:

Unit	Mains fuse (time-delay, ceramic, 5 × 20 mm, IEC 60127-2)
Zone	**T2A H 250 V** (1500 A breaking capacity)
4our	**T4A H 250 V** (1500 A breaking capacity)
8ight	**T8A 250 V** (1000 A breaking capacity)

- Never fit a higher rating, a glass or fast-blow substitute, or bridge a fuse.
- A fuse that blows repeatedly indicates a wiring or load fault — find and fix the cause; do not keep replacing the fuse.

Outputs. Each output is protected electronically (configurable current limit, microsecond short-circuit trip, over-temperature trip) **and** by an internal fuse inside the case. The internal fuses are not user-replaceable: if an output stays dead after a fault, contact JMB Lighting for service.

4. Load limits and heat

- Maximum per-output current: **6.75 A** on Zone units and 5 mΩ-sense modules, **6.5 A** on earlier 4our/8ight modules (configurable lower; set the limit to suit the connected strip — the app's current field shows the ceiling that applies). Total load must remain within the PSU rating.
- **Calibration controls (app → Setup → Developer mode) change what the protection measures.** The *Shunt* setting must match the fitted hardware — a wrong value halves or doubles every current reading and the trip point; leave it on *Auto* unless the unit has been modified. *Calibrate temperature* refuses any port more than 10 °C from ambient: that is a faulty sensor, and a fault must be repaired, never calibrated away. Calibrate current only against a trusted meter.
- Mount with free airflow. Do not cover the enclosure, stack heat-producing equipment on it, or enclose it without ventilation. Ambient operating range: 0–40 °C. **[Confirm final ambient rating before print.]**
- The unit reduces output automatically as fixtures approach their configured temperature limit and shuts a port down on a hard over-temperature; this protects the equipment and is not a substitute for correct installation.

5. Flashing-light effects — photosensitive epilepsy warning

Several built-in effects (Strobe, Lightning, Police, Paparazzi, Explosion and others) and DMX-driven strobe control produce **intense flashing light**. A small percentage of people may experience seizures when exposed to flashing light, even with no prior history.

- Warn cast, crew and audience before using strobe effects.
- **RDM discovery and Identify flash fixtures at full.** Any RDM console (or the app's RDM scan) running discovery makes every fixture it finds flash momentarily, and Identify flashes a fixture until switched off. This is the RDM standard and cannot be disabled on the fixture side. The app's *Flash to find* never lights a fixture: it uses the controller's own indicators or screens.
- Do not aim strobing fixtures at stairways, exits or moving-machinery areas.
- Stop use immediately if anyone experiences dizziness, disorientation or involuntary movement. **[LEGAL: standard PSE warning wording.]**

6. Infrared (IR) fixtures — invisible radiation

The controller can drive infrared LED strip and fixtures (Dim personality). **IR output is invisible: a fixture can be at full power while appearing dark.**

- Treat any connected IR fixture as ON whenever the system is powered.
- Do not stare into IR emitters at close range; position high-power IR sources so that prolonged close viewing is not possible. Fixture photobiological classification (IEC 62471) is the fixture manufacturer's responsibility — observe the fixture's own safety group and distances.
- Remember that camera monitoring shows IR that the eye cannot see — use a camera to confirm an IR rig is actually off before working close to it.

7. Radio equipment

The unit contains Bluetooth® Low Energy and, when fitted, a CRMX™ 2.4 GHz radio (CRMX by LumenRadio). Do not use where radio transmitters are prohibited. The radio link is not a safety-of-life channel: do not build safety functions (e.g. emergency blackout for egress lighting) solely on the wireless path. A CRMX receiver stays linked to its transmitter across power cycles; on wireless loss the controller **holds the last look** rather than going dark or starting an effect. Unlinking (app, or holding a unit's link button for over 3 seconds) makes the unit available to any transmitter in link mode — establish who owns the rig before unlinking on a working set. **[Compliance: Radio Equipment Regulations / RED assessment + Bluetooth SIG qualification to be completed before sale.]**

8. Installation practice

- Secure mounting only; do not suspend by cables.
- Strain-relieve all output wiring; chafed strip wiring is the most common cause of output faults.
- After installation, verify each port's current limit and temperature trip are set appropriately for the connected fixture, and function-test the fault clear path.
- The DMX/CRMX input, app and front panel all change live output. Establish who has control before working on fixtures — a port can light at any time while the system is powered. **Isolate power before touching wiring.**
- The app's **RDM scan takes over the output line** while it runs, so downstream fixtures lose their DMX for the duration; it releases the line when stopped or after 3 minutes idle. Do not start a scan during a take.
- **Firmware updates**: outputs go dark for a few seconds while a port module is updated and the controller restarts at the end. Update between setups, never during a live show.

9. Maintenance, service and modification

No user-serviceable parts inside any unit. Cleaning: dry or slightly damp cloth, mains disconnected, no solvents. Repairs only by JMB Lighting or its authorised agents; unauthorised opening, repair or modification ends the warranty **[LEGAL: warranty terms reference]** and may render the equipment unsafe.

10. Disposal

Do not dispose of in general waste. The unit is electrical and electronic equipment (WEEE): recycle through an appropriate collection scheme. **[Confirm WEEE producer registration + crossed-bin marking before sale.]**

11. Markings and compliance

[To complete before sale — tracked in the launch checklist:] UKCA/CE marking and Declaration of Conformity (Electrical Equipment (Safety) Regulations 2016 / LVD, EMC, RED, RoHS), Bluetooth SIG qualification, CRMX licensing acknowledgement, rating plate (voltage, frequency, power, fuse rating), and this guide's inclusion in the box or by QR link.

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